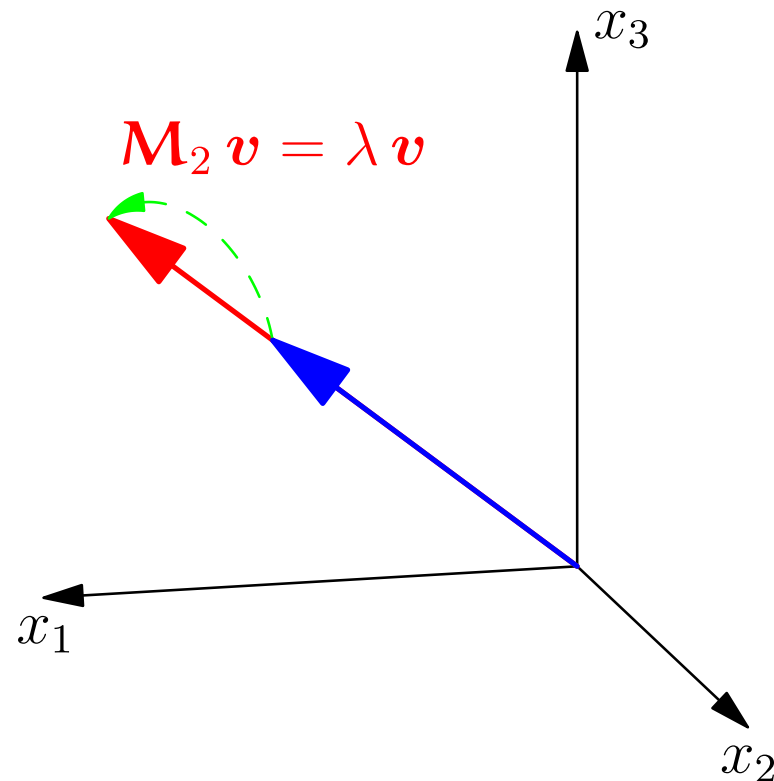


# Mathematics of Machine Learning

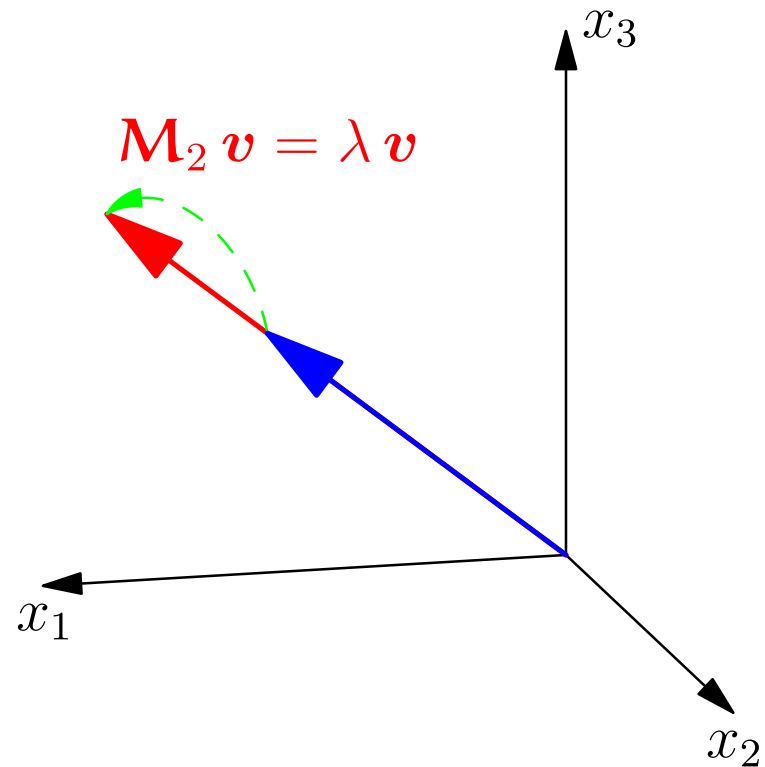
## *Eigensystems*



*Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank*

# Outline

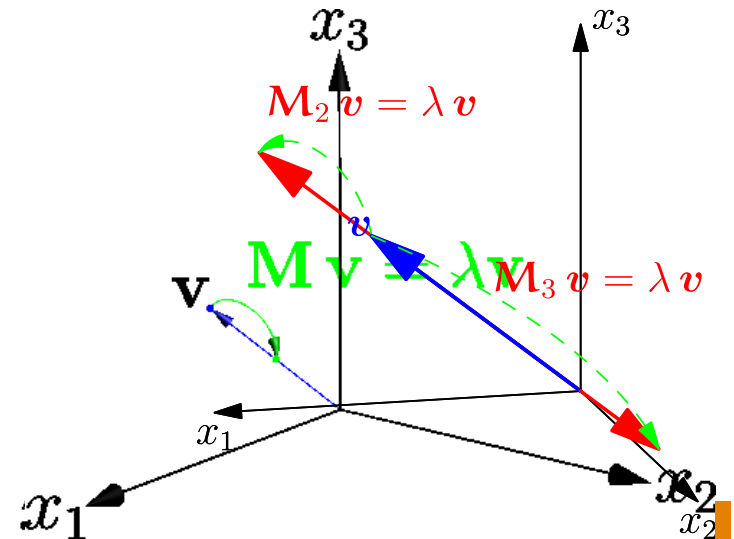
1. **Eigenvectors**
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation



# Eigenvector equation

- Eigen-systems help us to understand mappings
- A vector  $v$  is said to be an **eigenvector** if

$$Mv = \lambda v$$



- $M$  is square (i.e.  $n \times n$ )
- Where the number  $\lambda$  is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

# Symmetric Matrices

- If  $\mathbf{M}$  is an  $n \times n$  **symmetric** matrix then it has  $n$  real orthogonal eigenvectors with real eigenvalues■
- We denote the  $i^{th}$  eigenvector by  $\mathbf{v}_i$  and the corresponding eigenvalue by  $\lambda_i$  so that

$$\mathbf{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i \blacksquare$$

- Orthogonal means that if  $i \neq j$  then

$$\mathbf{v}_i^T \mathbf{v}_j = 0 \blacksquare$$

- (We can always normalise eigenvectors if we want)■

# Proof of Orthogonality

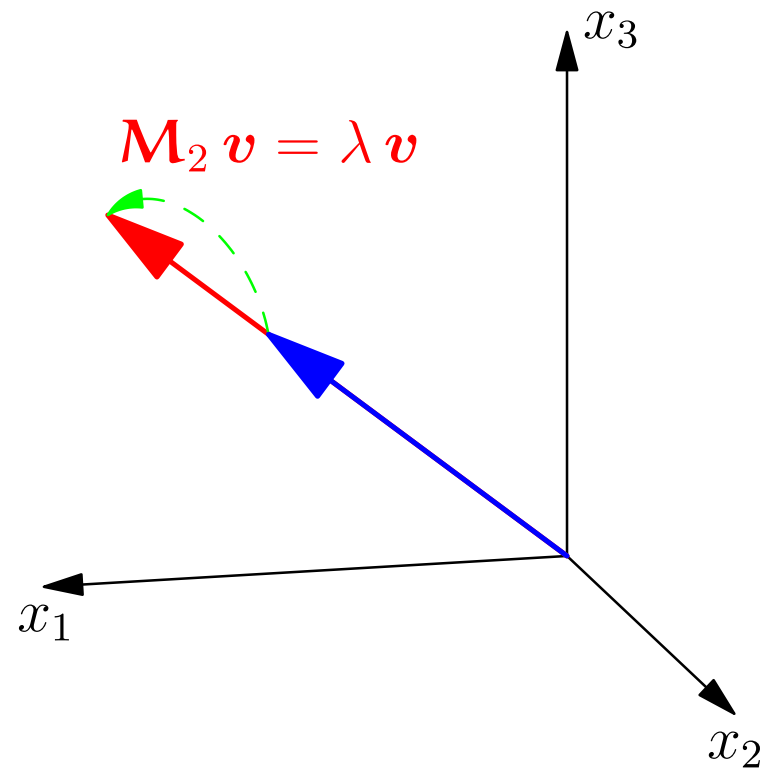
- $(\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i)^\top$  implies  $\mathbf{v}_i^\top \mathbf{M}^\top = \lambda_i\mathbf{v}_i^\top$  ■
- When  $\mathbf{M}$  is symmetric then  $\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i \Rightarrow \mathbf{v}_i^\top \mathbf{M} = \lambda_i\mathbf{v}_i^\top$  ■
- Consider two eigenvectors  $\mathbf{v}_i$  and  $\mathbf{v}_j$  of  $\mathbf{M}$

$$\begin{aligned}\mathbf{v}_i^\top \mathbf{M}\mathbf{v}_j &= (\mathbf{v}_i^\top \mathbf{M})\mathbf{v}_j = \lambda_i\mathbf{v}_i^\top \mathbf{v}_j \\ &= \mathbf{v}_i^\top (\mathbf{M}\mathbf{v}_j) = \lambda_j\mathbf{v}_i^\top \mathbf{v}_j\end{aligned}$$

- So either  $\lambda_i = \lambda_j$  or  $\mathbf{v}_i^\top \mathbf{v}_j = 0$  ■
- If  $\lambda_i = \lambda_j$  then any linear combination of  $\mathbf{v}_i$  and  $\mathbf{v}_j$  is an eigenvector  $(\mathbf{M}(a\mathbf{v}_i + b\mathbf{v}_j) = \lambda_i(a\mathbf{v}_i + b\mathbf{v}_j))$  ■ So I can choose two eigenvectors that are orthogonal to each other. ■

# Outline

1. Eigenvectors
2. **Orthogonal Matrices**
3. Eigen Decomposition
4. Low Rank Approximation



# Orthogonal Matrices

- We can construct an **orthogonal** matrix  $\mathbf{V}$  from the eigenvectors

$$\mathbf{V} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$$

- Matrix  $\mathbf{V}$  is an  $n \times n$  matrix
- Because of the orthogonality of the vectors  $\mathbf{v}_i$

$$\mathbf{V}^T \mathbf{V} = \begin{pmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \dots & \mathbf{v}_1^T \mathbf{v}_n \\ \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 & \dots & \mathbf{v}_2^T \mathbf{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_n^T \mathbf{v}_1 & \mathbf{v}_n^T \mathbf{v}_2 & \dots & \mathbf{v}_n^T \mathbf{v}_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = \mathbf{I}$$

# The Other Way Around

- We have shown that  $\mathbf{V}^T \mathbf{V} = \mathbf{I}$
- Thus multiply both sides on the left by  $\mathbf{V}$

$$\mathbf{V} \mathbf{V}^T \mathbf{V} = \mathbf{V}$$

- $\mathbf{V}$  will have an inverse,  $\mathbf{V}^{-1}$ , such that  $\mathbf{V} \mathbf{V}^{-1} = \mathbf{I}$
- Multiplying the equation on the right by  $\mathbf{V}^{-1}$

$$(\mathbf{V} \mathbf{V}^T) \mathbf{V} \mathbf{V}^{-1} = \mathbf{V} \mathbf{V}^{-1}$$

$$\mathbf{V} \mathbf{V}^T = \mathbf{I}$$

- Note that,  $\mathbf{V}^{-1} = \mathbf{V}^T$  (definition of orthogonal matrix)

# Invertible Matrices

- A matrix,  $\mathbf{M}$ , will be singular (uninvertible) if there exists a vector  $\mathbf{x} (\neq \mathbf{0})$  such that

$$\mathbf{M}\mathbf{x} = \mathbf{0}$$

- Now if there exists such a vector such that  $\mathbf{V}\mathbf{x} = \mathbf{0}$  then multiply by  $\mathbf{V}^T$  we get

$$\mathbf{V}^T\mathbf{V}\mathbf{x} = \mathbf{V}^T\mathbf{0}$$

$$\mathbf{x} = \mathbf{0}$$

since  $\mathbf{V}^T\mathbf{V} = \mathbf{I}$

- Thus  $\mathbf{V}$  is invertible

# Rotations

- Orthogonal matrices satisfy  $\mathbf{V}^T \mathbf{V} = \mathbf{V} \mathbf{V}^T = \mathbf{I}$
- As a consequent they define rotations (and possibly a reflection)
- Consider a vector  $x$  and  $x' = \mathbf{V}x$ , now

$$\|x'\|_2^2 = x'^T x' = (\mathbf{V}x)^T (\mathbf{V}x) = x^T \mathbf{V}^T \mathbf{V} x = x^T x = \|x\|_2^2$$

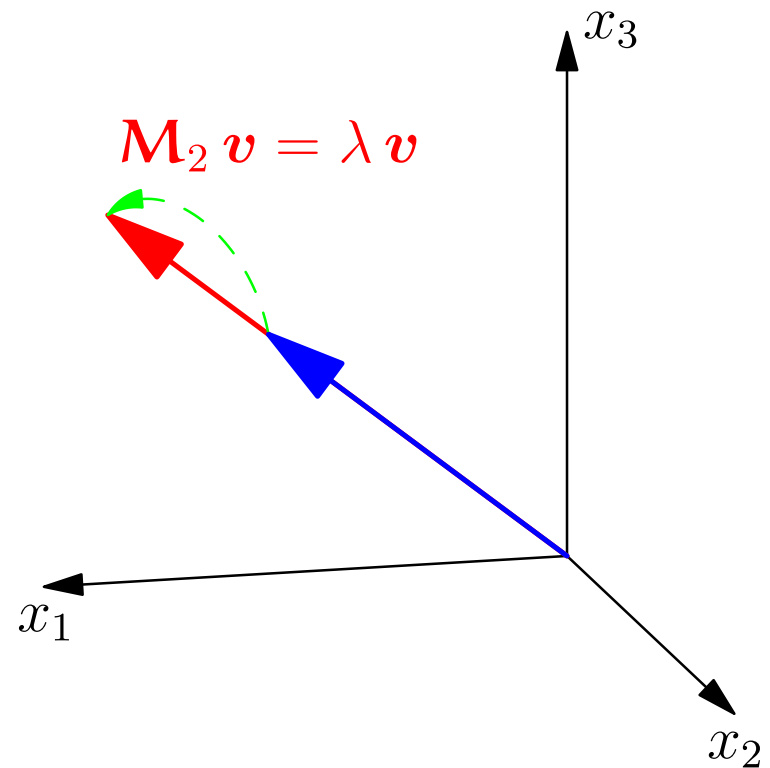
- Similarly if additionally  $y' = \mathbf{V}y$  then

$$\langle x', y' \rangle = (\mathbf{V}x)^T (\mathbf{V}y) = x^T \mathbf{V}^T \mathbf{V} y = x^T y = \langle x, y \rangle = \|x\|_2 \|y\|_2 \cos(\theta)$$

- Rotations and reflections preserve lengths and angles

# Outline

1. Eigenvectors
2. Orthogonal Matrices
3. **Eigen Decomposition**
4. Low Rank Approximation



# Matrix Decomposition

- Taking the matrix of eigenvectors,  $V$ , then

$$MV = M(v_1, v_2, \dots, v_n) = (\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n) = V\Lambda$$

- where  $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$

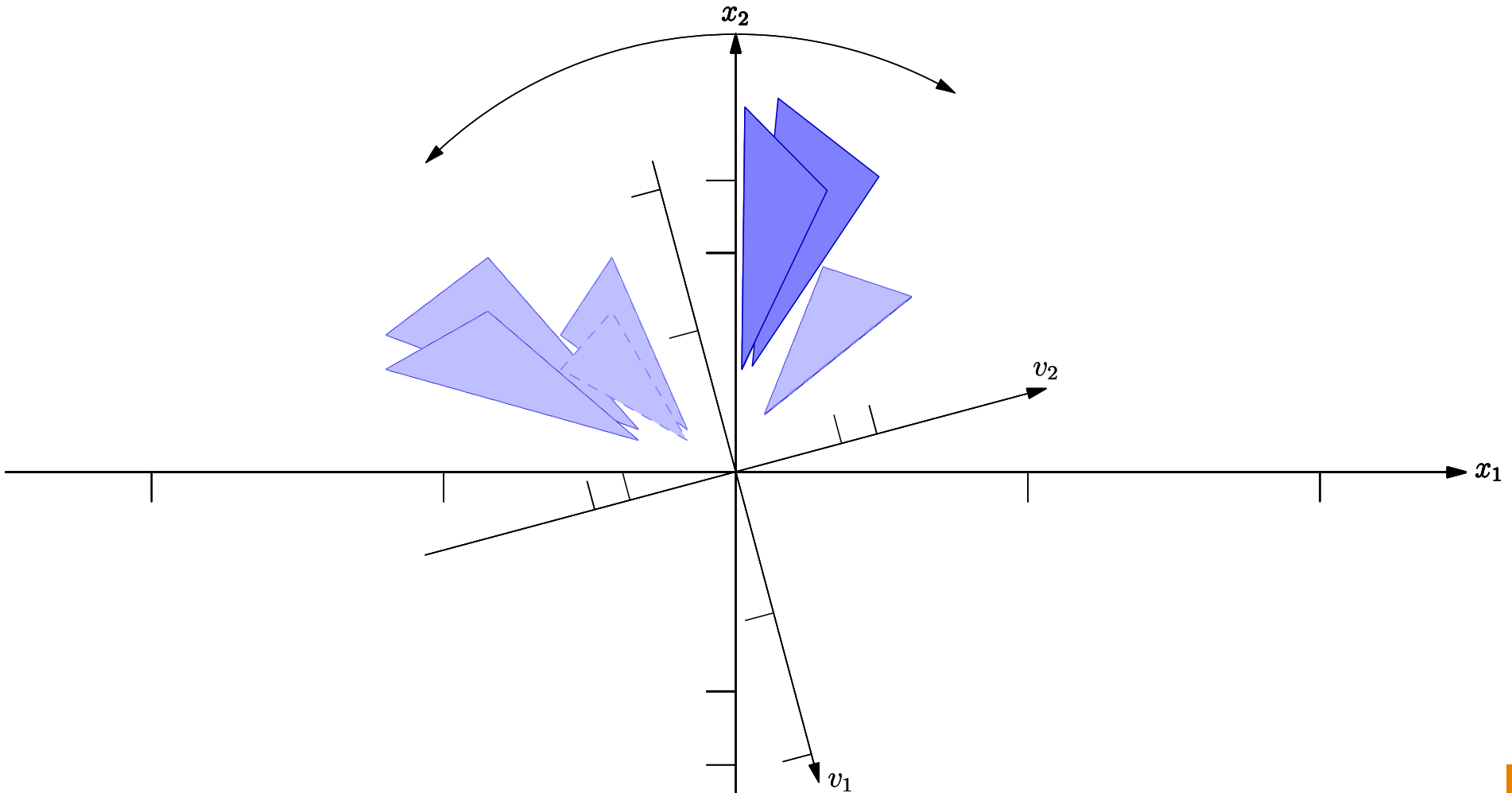
- Now

$$M = MVV^T = V\Lambda V^T$$

- Very important *similarity transform*

# Mappings by Symmetric Matrices

$$\mathbf{M} = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \begin{pmatrix} \cos(-75) & \sin(-75) \\ -\sin(-75) & \cos(-75) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75) & \sin(75) \\ -\sin(75) & \cos(75) \end{pmatrix}$$



# Inverses

- For any symmetric invertible matrix

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

$$\mathbf{M}^{-1} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T$$

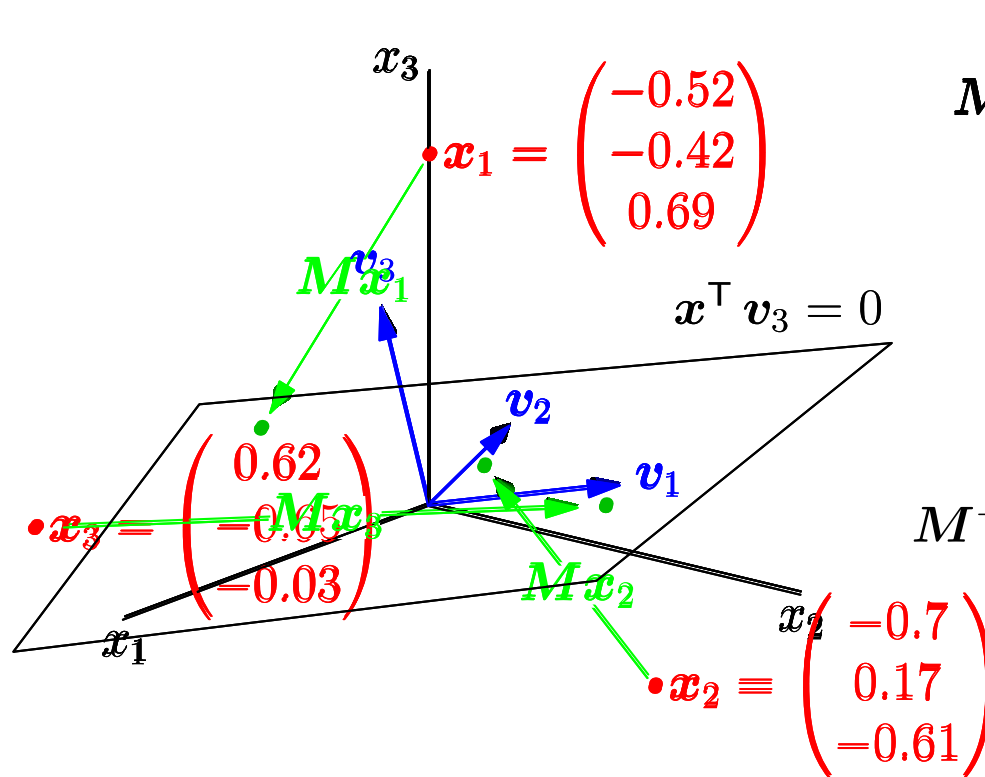
- Where  $\mathbf{\Lambda}^{-1} = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\lambda_n} \end{pmatrix}$

- Since

$$\begin{aligned} \mathbf{M}\mathbf{M}^{-1} &= (\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T)(\mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T) = \mathbf{V}\mathbf{\Lambda}(\mathbf{V}^T\mathbf{V})\mathbf{\Lambda}^{-1}\mathbf{V}^T \\ &= \mathbf{V}\mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{V}^T = \mathbf{V}\mathbf{V}^T = \mathbf{I} \end{aligned}$$

- I.e, Small eigenvalues become large eigenvalues and visa versa

# III-Conditioning Again



$$M = \begin{pmatrix} 0.80 & 0.01 & 0.25 \\ 0.01 & 0.85 & 0.48 \\ 0.25 & 0.48 & 0.5 \end{pmatrix} = V \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 1.25 & 0 \\ 0 & 0 & 0.1 \end{pmatrix} V^T$$

$$M^{-1} = \begin{pmatrix} 1.9 & 1.17 & -2.1 \\ 1.17 & 3.34 & -3.8 \\ -2.1 & -3.8 & 6.8 \end{pmatrix} = V \begin{pmatrix} 1.25 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 10 \end{pmatrix} V^T$$

# Condition Number

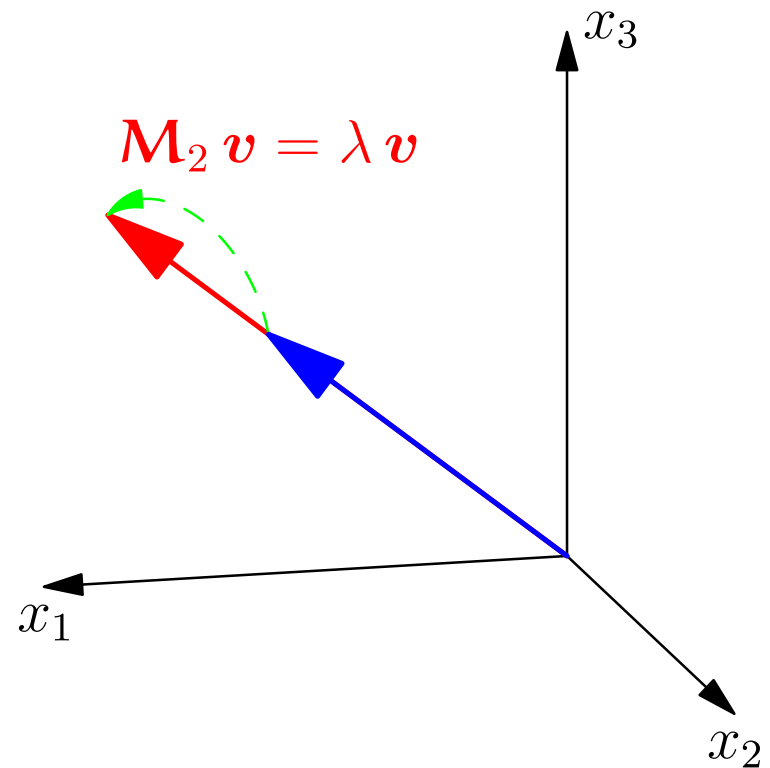
- Taking matrix inverses can be inherently unstable■
- Any small error can be amplified by taking the inverse■
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)■
- For invertible matrices we can take the largest eigenvalue as a norm of the matrix■
- The condition number is given by

$$\|\mathbf{M}\|_H \times \|\mathbf{M}^{-1}\|_H = \frac{|\lambda_{\max}|}{|\lambda_{\min}|} \blacksquare$$

- Large condition number implies very ill-conditioned■

# Outline

1. Eigenvectors
2. Orthogonal Matrices
3. Eigen Decomposition
4. **Low Rank Approximation**



# Rank of a Matrix

- The rank of a matrix,  $\mathbf{M}$ , is the number of non-zero eigenvalues■
- The space spanned by the eigenvectors  $\mathbf{v}_a$ ,  $\mathbf{v}_b$ , etc. with zero eigenvalue forms a **null space**■
- Any vector in the null space will get projected to the zero vector

$$\mathbf{M}(a\mathbf{v}_a + b\mathbf{v}_b + \dots) = \mathbf{0}■$$

- A square matrix is said to be **rank deficient** if it has any eigenvectors with eigenvalue equal to 0■
- This happens when the columns of the matrix are not linearly independent■

# “Inverting” Rank Deficient Matrices

- Rank deficient matrices are non-invertible (i.e. we don't know the vector  $x$  such that  $Mx = b$ ) as we don't know the component of the  $x$  in the null space■
- Although we don't know  $x$  we can find a vector,  $x$ , that satisfies  $Mx = b$ ■
- Given a symmetric  $n \times n$  matrix with  $k$  non-zero eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_k$  we can construct a “pseudo inverse”  $M^+$  as  $V\Lambda^+V^T$  where  $\Lambda^+ = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_k}, 0, \dots, 0)$ ■
- This finds the vector  $x$  with no component in the null space■(it is the solution with the smallest norm)■
- This is different to the pseudo inverse for non-square matrices■

# Low Rank Approximation

- Recall that matrices with large and small eigenvalues are ill-conditioned so the inverse has the potential to greatly amplify any measurement error■
- One work around is to set all small eigenvalues to zero and use the pseudo inverse■
- Setting small eigenvalues to zero reduces the rank of the matrix and is an example of a low rank approximation■
- Low rank approximations are much used to obtain approximate models for arrays of data■ (we will revisit this when we look at SVD)■

# Summary

- Linear mappings are commonly used in machine learning algorithms such as regression■
- We can understand symmetric operators by looking at their eigenvectors■
- Any symmetric matrix can be decomposed as  $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$ 
  - ★ where  $\mathbf{V}$  are orthogonal matrices whose rows are the eigenvector
  - ★ and  $\mathbf{\Lambda}$  is a diagonal matrix of the eigenvalues■
- This decomposition allows us to understand inverse mappings■